

THE SKYLAB

SKYLAB WAS THE FIRST AMERICAN space station. Coming at a time when NASA was under pressure over its budget, the emphasis was on economy and practicality. Creating a space station by adapting the third stage of a Saturn 5 rocket was much cheaper than starting from scratch, and to emphasize the practicality of the project it was dubbed an Orbital Workshop. There were to be medical tests of the effects of weightlessness, observation of the Sun and stars, photography of the Earth's surface, and experiments concerning possible industrial activities in space.

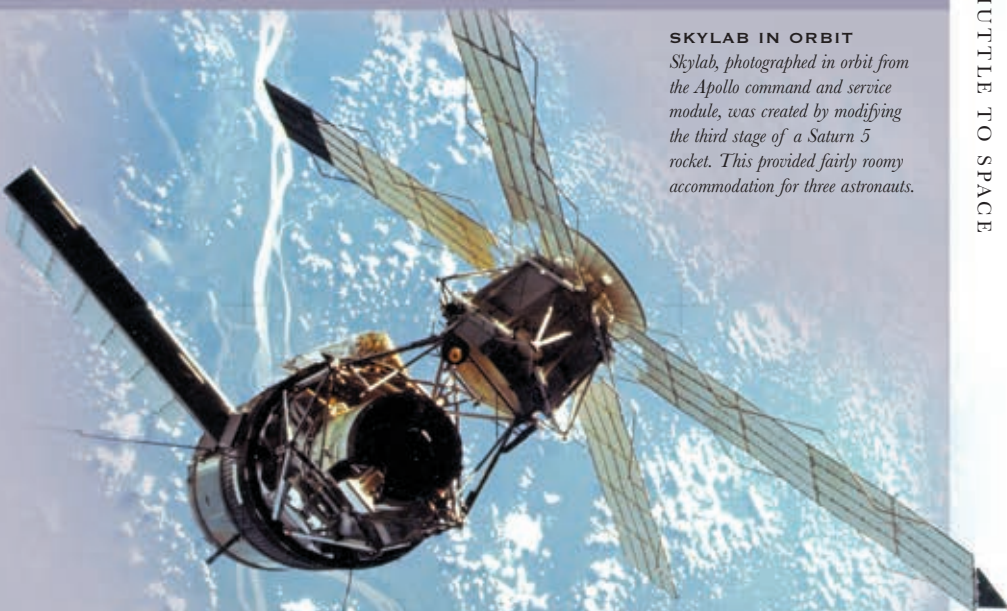
Skylab was launched into orbit on May 14, 1973, the crew following 11 days later in an Apollo module. The project began with a crisis. An accident during the ascent left Skylab with no thermal shield and only one of its two solar panels operational. The crew – Charles Conrad, Paul Weitz, and Joseph Kerwin – were fortunately able to repair the damage to the rapidly overheating craft by passing a sunshade through an airlock.

The initial crew stayed on Skylab for 28 days, but two subsequent three-man crews stayed for 59 days and 84 days respectively. Skylab was about as spacious as a small house, with three sleeping areas, a kitchen and eating space, a bathroom with a gravity-free air-suction toilet, and a shower – which had to be an enclosed container, to stop the water from floating away. The astronauts were committed to an intensive program of experiments and medical tests. Some of the effects of weightlessness were strange – for example, the Skylab crew members grew

in height by about an inch during their stay on board. Conrad quipped: "At last I'm taller than my wife." Constant exercise maintained muscle tone, which would otherwise have tended to deteriorate. Although they complained about relentless work schedules and tasteless food, the astronauts had the sensational ever-changing view of the Earth to cheer their spirits. In this they were somewhat fortunate, as Saturn engineers had originally intended not to put in a window.

SKYLAB IN ORBIT

Skylab, photographed in orbit from the Apollo command and service module, was created by modifying the third stage of a Saturn 5 rocket. This provided fairly roomy accommodation for three astronauts.



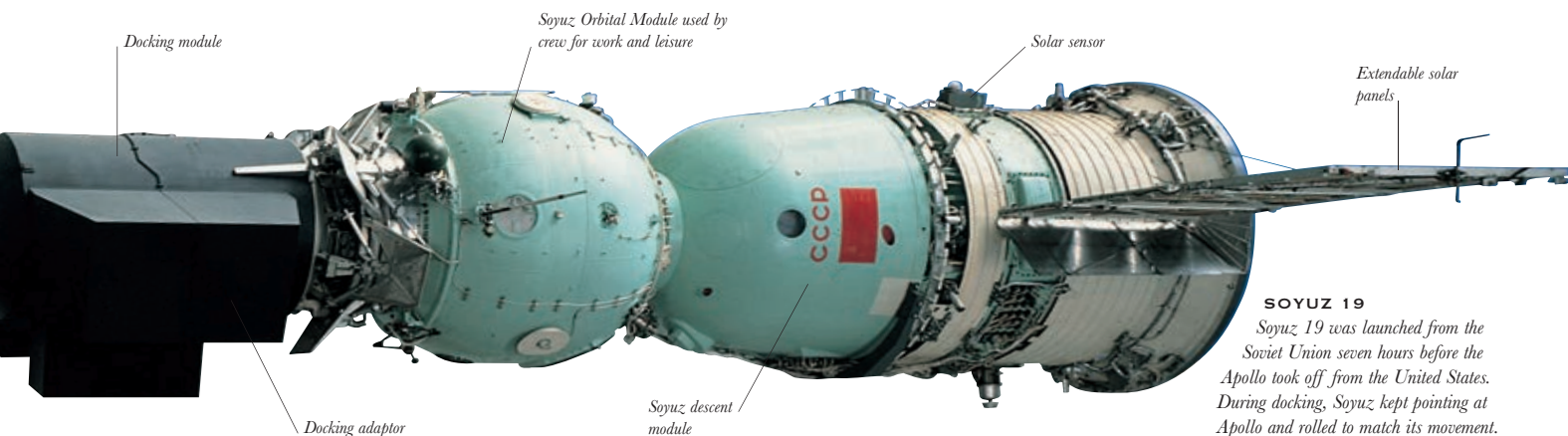
INSIDE SKYLAB

Skylab had a kitchen and dining area, and facilities for medical tests. Astronauts found that prolonged weightlessness had little adverse effect on health, apart from an embarrassing excess of gas after eating.



SKYLAB BADGE

The Skylab project was rated as a major success. The spacecraft remained in orbit for six years, although astronauts were aboard for only a total of 171 days.



SOYUZ 19

Soyuz 19 was launched from the Soviet Union seven hours before the Apollo took off from the United States. During docking, Soyuz kept pointing at Apollo and rolled to match its movement.